

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I Shaik Mubarak(192425194)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Shaik Mubarak

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.

TITLE: Generative AI Shopping Assistant for Personalized Retail Experience

AI Domain: Retail

1. Domain Selection

The selected application belongs to the **Retail** domain. Retail businesses use Artificial Intelligence to improve customer satisfaction, automate shopping assistance, provide personalized recommendations, manage inventory, and increase sales. AI helps retailers understand customer preferences and deliver better shopping experiences through intelligent decision-making.

2. Introduction and Background

Artificial Intelligence (AI) is transforming the retail industry by providing intelligent and personalized shopping experiences. One of the latest innovations is the **Generative AI Shopping Assistant**, which acts as a virtual shopping companion for customers. It understands natural language queries, recommends suitable products, compares items, answers customer questions, and assists throughout the shopping process.

Unlike traditional chatbots, Generative AI Shopping Assistants generate human-like responses using Large Language Models (LLMs). They analyze customer preferences, purchase history, and browsing behavior to provide personalized recommendations. Major companies such as Amazon, Walmart, Flipkart, and Shopify have integrated AI shopping assistants to improve customer engagement and business performance.

3. Problem Identification

Traditional retail systems face several challenges:

- Customers spend significant time searching for suitable products.
- Generic product recommendations reduce customer satisfaction.
- Manual customer support is expensive and time-consuming.
- Large product catalogs make decision-making difficult.
- Businesses struggle to understand changing customer preferences.
- Poor personalization leads to lower sales and customer retention.

These challenges reduce shopping efficiency and negatively impact both customers and businesses.

4. Need for AI Solution

Artificial Intelligence is required because traditional recommendation systems provide limited personalization and cannot understand natural language effectively.

AI improves traditional retail systems by:

- Providing personalized product recommendations.
- Understanding customer queries using Natural Language Processing.

- Offering instant customer support.
- Learning customer preferences using Machine Learning.
- Predicting customer behavior and future purchases.
- Improving inventory planning through predictive analytics.
- Reducing operational costs by automating customer interactions.

As a result, AI enhances both customer satisfaction and business productivity.

5. AI Technologies Used

The Generative AI Shopping Assistant combines multiple AI technologies.

Machine Learning (ML)

Learns customer purchasing behavior and continuously improves recommendation accuracy.

Deep Learning

Processes large volumes of customer data and recognizes complex shopping patterns.

Natural Language Processing (NLP)

Understands customer questions and generates human-like responses.

Generative AI

Uses Large Language Models to generate personalized conversations, explain products, and assist customers during shopping.

Recommendation Engine

Suggests products based on customer interests using collaborative filtering and content-based filtering techniques.

Predictive Analytics

Forecasts customer demand and predicts future buying trends.

6. Working Mechanism

The AI Shopping Assistant follows these steps:

1. Customer enters a shopping query through a website or mobile application.
2. Natural Language Processing analyzes the query.
3. Machine Learning identifies customer preferences.
4. The Recommendation Engine searches suitable products.
5. Generative AI creates personalized responses.
6. The chatbot displays recommended products.
7. Customer selects and purchases the preferred product.

Workflow



7. Data Sources and Processing

The AI system collects data from multiple sources.

Data Sources

- Customer purchase history
- Browsing history
- Product reviews
- Customer ratings
- Shopping cart information
- Product catalog
- Inventory database
- Customer feedback

Data Processing

- Data collection
- Data cleaning
- Feature extraction
- Customer behavior analysis
- Recommendation generation
- Personalized response generation

The processed data enables intelligent and accurate decision-making.

8. Real-World Implementation

Several leading organizations have successfully implemented Generative AI Shopping Assistants.

Organization	AI Application
Amazon	AI Shopping Assistant & Personalized Recommendations
Walmart	AI Customer Support & Inventory Prediction
Flipkart	Personalized Product Recommendation
Shopify	AI Shopping Assistant
Sephora	AI Beauty Recommendation Assistant

IKEA	AI Interior Design Shopping Assistant
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These companies use AI to improve customer experience and increase sales.

9. Impact and Benefits

The implementation of Generative AI Shopping Assistants provides significant benefits.

For Customers

- Personalized shopping experience
- Faster product search
- Better recommendations
- Instant customer support
- 24×7 availability
- Easy product comparison

For Businesses

- Increased sales
- Higher customer satisfaction
- Improved customer retention
- Reduced operational costs
- Better inventory management
- Improved marketing strategies
- Increased business efficiency

Overall, AI enhances both customer experience and business profitability.

10. Challenges and Limitations

Despite its advantages, the AI Shopping Assistant faces several challenges.

Technical Challenges

- Large computational requirements
- Continuous model training
- Data quality issues

Privacy Challenges

- Collection of customer personal information
- Protection of user data
- Compliance with privacy regulations

Security Challenges

- Cybersecurity threats
- Data breaches
- Unauthorized access

Ethical Challenges

- Bias in AI recommendations
- Lack of transparency
- Over-dependence on AI

Deployment Challenges

- High implementation cost
- Integration with existing systems
- Regular maintenance and updates

11. Future Scope and Emerging Trends

Generative AI Shopping Assistants will continue to evolve with advanced technologies.

Future developments include:

- AI-powered virtual shopping companions
- Voice-enabled shopping assistants
- Augmented Reality (AR) virtual try-on systems
- Virtual Reality (VR) shopping experiences
- Emotion-aware AI assistants
- Hyper-personalized recommendations
- Smart retail stores with autonomous checkout
- Integration with IoT devices
- Real-time multilingual support

Future research aims to improve recommendation accuracy, privacy protection, and conversational intelligence.

12. Conclusion

Generative AI Shopping Assistants represent one of the most advanced applications of Artificial Intelligence in the retail sector. By combining Machine Learning, Natural Language Processing, Deep Learning, Recommendation Systems, and Generative AI, these intelligent systems provide personalized shopping experiences, improve customer satisfaction, and increase business efficiency. As AI technology continues to evolve, Generative AI Shopping Assistants will become an essential component of future retail systems, enabling smarter, faster, and more intelligent shopping experiences.

13. References

1. OpenAI. *Generative AI and Large Language Models*. <https://openai.com>
2. Amazon Science. *AI in Retail and Recommendation Systems*. <https://www.amazon.science>
3. Google Cloud. *Generative AI for Retail*. <https://cloud.google.com/solutions/retail>
4. Microsoft AI Blog. *Generative AI Applications in Retail*. <https://www.microsoft.com/ai>
5. McKinsey & Company. *The State of AI in Retail*. <https://www.mckinsey.com>
6. IBM Research. *Artificial Intelligence in Retail Industry*. <https://research.ibm.com>
7. Gartner Research. *Emerging Trends in AI-Powered Retail*. <https://www.gartner.com>

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (59–50%)
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.